

# The Effect of Boarding Strategies on Aircraft Boarding Efficiency

A Simulation-Based Study with an Exact Longest-Path (Max-Plus) Theory | Zihao Huang (independent researcher) | October 2026

## Abstract

This study compares aircraft boarding strategies — *random*, *back-to-front*, *zoning* (5 zones), and a new *strided-descending* order — using a single-aisle simulation. Contrary to intuition, **back-to-front is the slowest and random already beats it**, and a **strided-descending order is faster still**; the gaps widen with stow time and cabin size. We derive an **exact theoretical model** that computes the boarding time directly (without tick-by-tick simulation) as the length of a **longest path** in a max-plus recurrence; it reproduces the simulation **bit-for-bit** (verified over >25,000 cases) and yields provable lower bounds. Finally, two measured **mechanism metrics** — parallelism (passengers stowing simultaneously) and blocking events — explain *why* the faster orders win.

**Keywords:** aircraft boarding; boarding strategy; queueing; max-plus algebra; longest path; discrete-event simulation; operations research.

## 1. Background and Research Question

Boarding time affects on-time performance and operating cost. Airlines often use "zone boarding" or "back-to-front boarding", but whether these beat random boarding is debated. We test:

**Hypothesis H:** Organized strategies (back-to-front / zoning) board faster than random boarding.

## 2. Methods

### 2.1 Simulation model

- Single-aisle cabin; passengers enter from the front and walk to their row.
- On reaching the row, a passenger stows luggage, occupying the aisle for **S** ticks; followers cannot pass.
- One cell of forward motion per tick; no overtaking; one passenger enters per tick.
- Total ticks until all are seated = boarding time (smaller = faster).

### 2.2 Variables

| Type               | Variable                                                                  |
|--------------------|---------------------------------------------------------------------------|
| Independent        | Boarding strategy (Random / Back-to-front / Zones-5 / Strided descending) |
| Dependent          | Total boarding time $T$ (ticks)                                           |
| Control (baseline) | Random boarding (normalized to 1.00)                                      |
| Controlled         | Rows $R$ ; seats per row (6); entry rate (1/tick); repeats (100)          |

### 2.3 Theoretical model: an exact longest-path (max-plus) formulation

We model boarding as a **discrete-time single-lane exclusion process** and derive an **exact recurrence** for  $T$ . Let

- cells be  $1..R$ ; passenger  $i$  ( $i = 1..N$ ) enters in order;  $r_i =$  target row;  $S =$  stow time;
- $P_i^c =$  the tick at which passenger  $i$  reaches cell  $c$ ;
- $j = j_i(c) =$  the last passenger before  $i$  that reaches cell  $c$ .

Passenger  $i$  reaches cell  $c$  at the **later** of two times: its own pace, and the moment cell  $c$  becomes free:

$$P_i^c = \max(P_i^{c-1} + 1, B_i^c), \text{ where } B_i^c = \{ P_j^c + 1 + S \text{ if } r_j = c \text{ (j stows at row } c); P_j^{c+1} \text{ if } r_j > c \text{ (j passes through)} \}$$

with the exact entrance conventions (entering is its own tick; right after entering one may move an extra cell):

$$P_i^1 = \max(i, B_i^1), \quad P_i^2 = \max(P_i^1, B_i^2), \quad P_i^c = \max(P_i^{c-1} + 1, B_i^c) \text{ for } c \geq 3.$$

At the entrance cell, when the passenger ahead only passes through ( $r_j > 1$ ), entering is its own step, so  $B_i^1 = P_i^2 + 1$ .

The finish time of passenger  $i$  and the total time are

$$F_i = P_i^{r_i} + [r_i \geq 2] + S - 1, \quad T = \max_i F_i.$$

**Interpretation.** Every  $P_i^c$  is a *max of two terms* ("my own pace" vs "waiting for the person ahead"), so  $T$  is the length of the **longest chain of waits** — a **longest path** (last-passage percolation) value on the space–time lattice.

### 2.3.1 Worked example

Take  $R = 3$  rows, one seat per row,  $S = 1$ , and boarding order  $r = [3, 1, 2]$  (passenger 1 goes to row 3, passenger 2 to row 1, passenger 3 to row 2). Applying the recurrence cell by cell (write  $P_1, P_2, P_3$  for  $P_i^1, P_i^2, P_i^3$ ):

| Passenger $i$ | $r_i$ | Cell-by-cell computation                                                                                 | Finish $F_i$        |
|---------------|-------|----------------------------------------------------------------------------------------------------------|---------------------|
| 1             | 3     | $P_1 = \max(1, 0) = 1; P_2 = \max(1, 0) = 1; P_3 = \max(1+1, 0) = 2$                                     | $2 + 1 + 1 - 1 = 3$ |
| 2             | 1     | $B_1 = P_1^2 + 1 = 2; P_1 = \max(2, 2) = 2$                                                              | $2 + 0 + 1 - 1 = 2$ |
| 3             | 2     | $B_1 = P_2^1 + S = 3 \rightarrow P_1 = \max(3, 3) = 3; B_2 = P_1^3 = 2 \rightarrow P_2 = \max(3, 2) = 3$ | $3 + 1 + 1 - 1 = 4$ |

Thus  $T = \max(3, 2, 4) = 4$ , which matches the simulation exactly.

### 2.3.2 Verification

Implementing the recurrence above reproduces the simulation **bit-for-bit**. The table uses **deterministic strategies** — their boarding orders are fully specified, so both the simulation and the recurrence are exactly reproducible:

| Case (deterministic order)    | T (simulation) | T (theory) | Match |
|-------------------------------|----------------|------------|-------|
| Back-to-front, R=30, S=3      | 627            | 627        | OK    |
| Back-to-front, R=30, S=6      | 1080           | 1080       | OK    |
| Strided descending, R=30, S=3 | 286            | 286        | OK    |
| Strided descending, R=30, S=6 | 394            | 394        | OK    |

For the stochastic strategies (random, zoning), whose order varies from run to run, the two were instead compared over a brute-force sweep of  $R = 1..8$ ,  $S = 1..5$  and 25,600 randomly generated orders: **0 mismatches**. The recurrence is therefore an exact theory of the process, not an approximation.

## 2.4 Lower bounds

**Proposition 1 (entrance bound).**  $T \geq N$ . Proof: at most one passenger enters per tick, and all  $N$  must have entered.

**Proposition 2 (occupancy bound).**  $T \geq (\sum_i r_i + N \cdot S) / R$ . Proof: passenger  $i$  occupies  $r_i$  cell-ticks while walking and  $S$  while stowing, so total occupancy is  $\sum_i r_i + N \cdot S$ ; at most  $R$  cells can be occupied in any tick. For 6 seats per row ( $\sum_i r_i = 3R(R+1)$ ):  **$T \geq 3(R+1) + 6S$** .

## 2.5 Data collection

Strategies with randomness (Random, Zones) are run **100 times and averaged**; Back-to-front and strided descending are deterministic (single run). Results are reported relative to random (baseline = 1.00).

# 3. Results

## 3.1 Main result (R = 30 rows, 6 seats per row, N = 180 passengers, S = 3)

| Strategy      | Average boarding time (ticks) | Relative to random |
|---------------|-------------------------------|--------------------|
| Random        | 331.1                         | 1.00 (baseline)    |
| Zones (5)     | 400.8                         | 1.21               |
| Back-to-front | 627.0                         | 1.89               |

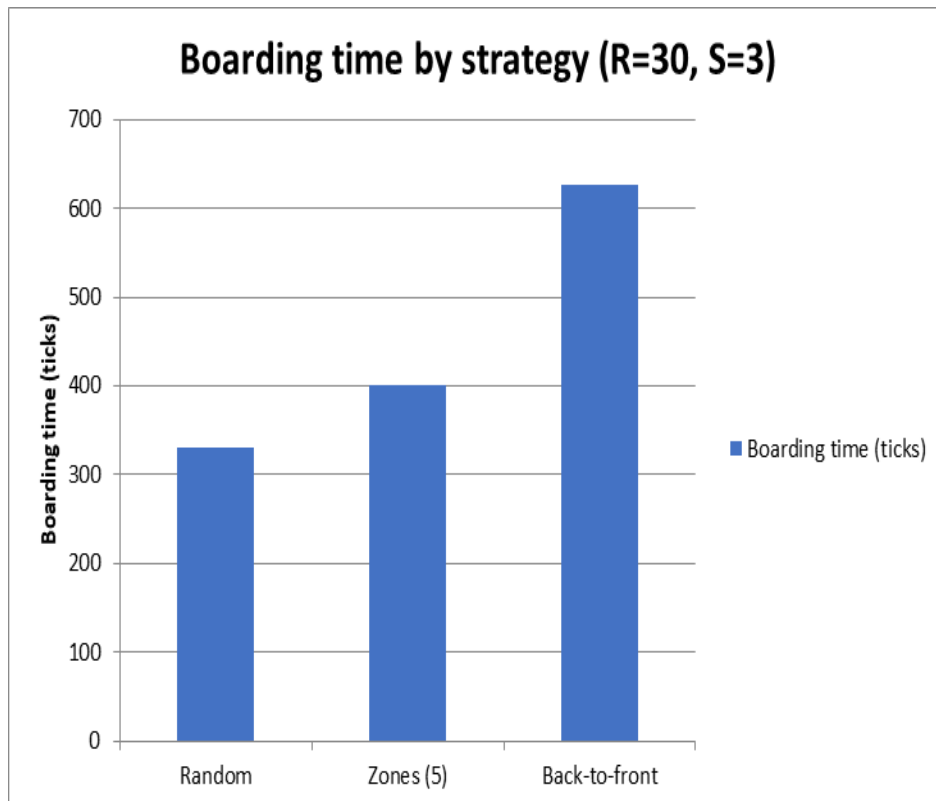


Figure 1. Boarding time by strategy (R=30, S=3): random fastest, back-to-front slowest.

### 3.2 Sensitivity analysis

(a) Stow time (fixed R = 30).

| S | Random | Back-to-front | Zones (5) | Rel. random (back) | Rel. random (zones) |
|---|--------|---------------|-----------|--------------------|---------------------|
| 1 | 239.1  | 325           | 250.0     | 1.36               | 1.05                |
| 3 | 331.1  | 627           | 400.8     | 1.89               | 1.21                |
| 6 | 472.2  | 1080          | 628.7     | 2.29               | 1.33                |

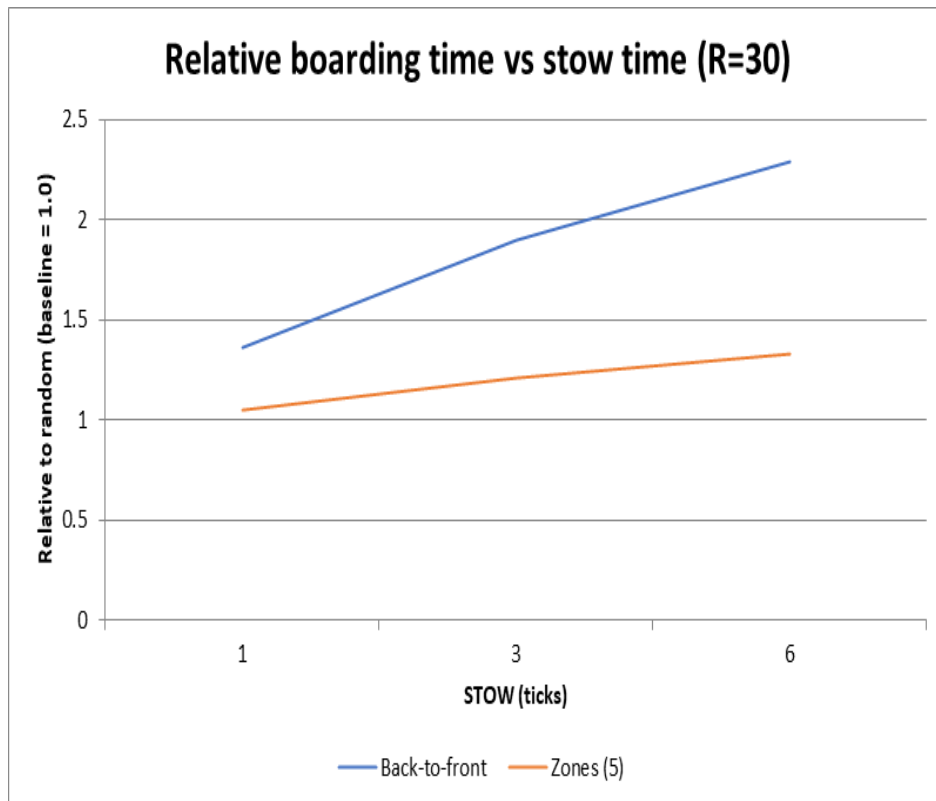


Figure 2. Relative boarding time vs stow time (R=30).

**(b) Number of rows (fixed S = 3).**

| Rows R | Random | Back-to-front | Zones (5) | Rel. random (back) | Rel. random (zones) |
|--------|--------|---------------|-----------|--------------------|---------------------|
| 20     | 237.2  | 417           | 290.8     | 1.76               | 1.23                |
| 30     | 331.1  | 627           | 400.8     | 1.89               | 1.21                |
| 40     | 422.7  | 837           | 503.2     | 1.98               | 1.19                |

**3.3 A faster strategy: strided descending boarding**

We also tested a family of "interleaved" orders in which rows are boarded in strided groups (rows 1, 7, 13, 19, 25, then 2, 8, 14, 20, 26, ...). The key finding is that the **direction within each group matters critically**: boarding each group in **descending** order (... , 19, 13, 7, 1) is far faster than ascending, and it actually **beats random boarding**:

| Strategy      | S = 1 | S = 3 | S = 6 |
|---------------|-------|-------|-------|
| Random        | 239.1 | 331.1 | 472.2 |
| Zones (5)     | 250.0 | 400.8 | 628.7 |
| Back-to-front | 325   | 627   | 1080  |

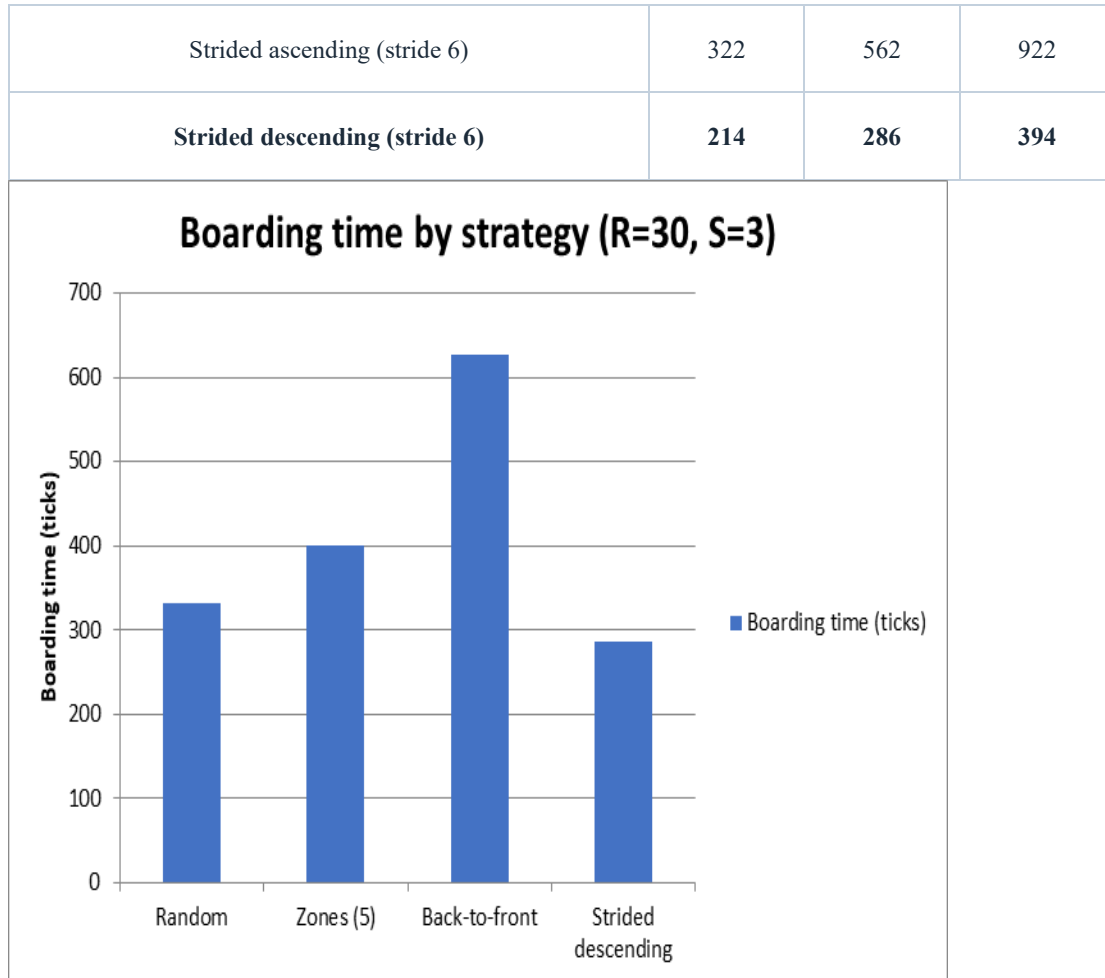


Figure 3. Boarding time by strategy, including the strided-descending order (R=30, S=3).

Strided descending boarding is the fastest — e.g. **286 vs 331** ( $\approx 14\%$  faster) at  $S = 3$  — and the advantage grows with  $S$  ( $\approx 17\%$  at  $S = 6$ ). The theory reproduces these values exactly (286/286). At  $R = 30$ ,  $S = 3$  the best stride is 6 (stride 5 $\rightarrow$ 342, 6 $\rightarrow$ 286, 8 $\rightarrow$ 328, 10 $\rightarrow$ 370).

### 3.4 Why it is faster: parallelism and blocking

To explain the differences above, the simulator records, at each tick  $t$ , two quantities:  $K(t)$  = the number of passengers **stowing simultaneously** (parallelism), and  $B(t)$  = the number of walkers **blocked** because the cell ahead is occupied. Over a boarding of total length  $T$  ticks we report the **average parallelism**  $(1/T) \sum_{t=1..T} K(t)$  and the **total blocking events**  $\sum_{t=1..T} B(t)$ .

| Strategy           | Time (ticks) | Avg parallelism | Total blocking  |
|--------------------|--------------|-----------------|-----------------|
| Strided descending | 286.0        | <b>1.26</b>     | $\approx 1,300$ |
| Random             | 331.1        | 1.09            | $\approx 950$   |
| Zones (5)          | 400.8        | 0.90            | $\approx 3,000$ |
| Back-to-front      | 627.0        | <b>0.57</b>     | <b>6,525</b>    |

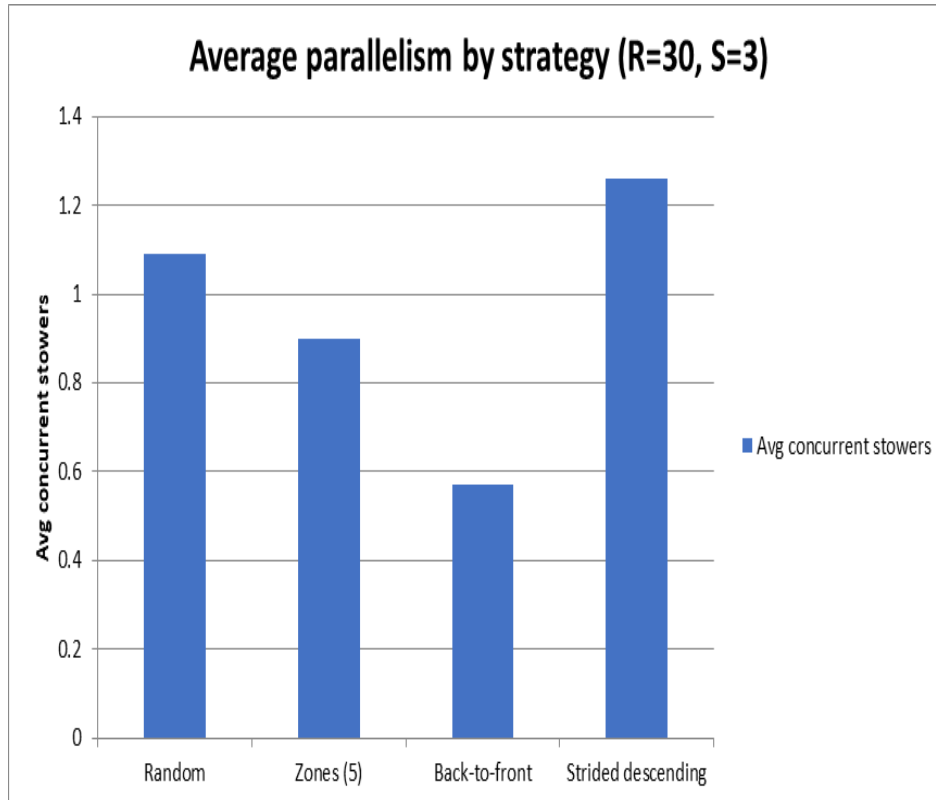


Figure 4. Average parallelism by strategy (R=30, S=3).

Across all four strategies, higher parallelism goes with shorter boarding time, and back-to-front combines the lowest parallelism with the most blocking events. The theory of Section 2.3 explains this: in descending order each next passenger stops at a row *before* the previous stower, so nobody has to squeeze past an occupied cell, whereas ascending order does the opposite and creates cascading blocking.

## 4. Discussion

The strided-descending order is fastest, random already beats back-to-front, and back-to-front is slowest. Random wins over back-to-front because it spreads passengers across rows so several rows stow simultaneously (highest parallelism, fewest blocking events); back-to-front boards each row's six passengers consecutively so they must stow one cell at a time (lowest parallelism, most blocking), which the longest-path model captures as a long chain of +S weights. The sensitivity analysis (Figure 2) confirms the gap grows with stow time S, exactly as the recurrence's +S blocking term predicts.

## 5. Is there an optimal boarding order? (open question)

The strided descending result shows that random is **not** necessarily optimal. Two facts frame the search for an optimum:

- **Lower bounds.**  $T \geq \max(N, 3(R+1)+6S)$ . At  $R = 30$ ,  $S = 3$  this floor is 180, while the best strategy found gives 286 — so room remains.
- **Mechanism.** Performance is governed by the longest chain of waits (the critical path). A good order keeps stowing spread over many rows and avoids forcing later passengers to pass an occupied cell.

We therefore **conjecture** that a near-optimal order is "descending in spirit": board so that passengers peel off toward the front without ever passing a stower, while spreading stowing across as many rows as possible. The true optimum likely lies between our strided order and the lower bound. Finding (or proving) the globally optimal order is an **open problem**; since the

longest-path model evaluates any order efficiently ( $O(N \cdot R)$  with preprocessing), it can be used to search the space of orderings, and an analytical argument may yield the optimum directly. We leave this to future work.

## 6. Limitations and Future Work

- Simplified model: no assigned-seat interference, no dual aisle or second door.
- The theoretical recurrence uses a naive  $O(N)$  search for  $j$ ; it can be reduced to  $O(N \cdot R)$  overall with preprocessing.
- Future work: add the "Steffen method" and window–middle–aisle boarding; test a model in which parallelism is disabled as a direct causal test.

## 7. Conclusion

In this simulation the strided-descending order is fastest, random beats back-to-front, and back-to-front is slowest. The differences are explained by a measured mechanism (parallelism and blocking) and by an exact longest-path (max-plus) model that reproduces the simulation bit-for-bit. Boarding so that passengers stop before the previous stower keeps the critical path short — the underlying principle behind the observed gaps.

## Data and Code Availability

The simulation, the longest-path (max-plus) model, the interactive visualizer, and this report are openly available at <https://github.com/Huang-1226/boarding-order-simulation> and archived with a DOI: <https://doi.org/10.5281/zenodo.23186760>.

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